

Mohammed Azharudeen Farook Deen

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About

Robotics Engineer with a strong focus on melding **deep learning**, **reinforcement learning**, and **optimal control** to create agile and collaborative robots.

Relevant Work Experience

Intern

Fraunhofer IPA

Stuttgart, Germany

July 2024 – Mar 2025

- Developed a custom simulation environment in **Mujoco** with both state-based and vision-based learning for a peg-in-hole task with different difficulty presets.
- Adapted the simulation environment to work with real-time Reinforcement Learning (SAC) and Imitation Learning (BC, HG-DAGger) pipelines in both **Pytorch** and **JAX**; Implemented **ROS1** / **ROS2** frameworks to control real robot (UR5e + Robotiq Hand-e gripper).
- Implemented and tuned various controller modules; Integrated teleoperation modules for keyboard and spacomouse control for **Learning from Demonstrations** and **Human-in-the-Loop RL**.

Master Thesis - Robot Learning of Quadrupedal Locomotion on Deformable Terrain

Aachen, Germany

Sep 2023 – Mar 2024

RWTH Institute for Data Science in Mechanical Engineering (DSME) and RWTH Institute of Geomechanics and Underground Technology (GUT)

- Integrated a particle-based contact model to simulate the complex behaviors of granular materials in **NVIDIA Isaac Sim**, developed an adaptable and robust controller using reinforcement learning (PPO) in PyTorch for quadruped (Unitree A1) to follow a unified policy in diverse natural environments and evaluated the performance with sim-to-real additions.
- Preliminary work accepted at the 1st German Robotics Conference (GRC2025), and a related paper is being finalized for publication in the journal

Student Assistant

Aachen, Germany

RWTH Institute for Automatic Control (RWTH IRT)

May 2023 – June 2024

- Extended codebase of a unified C++ interface for various **quadratic programming solvers(QP)** to **nonlinear programming(NLP) solvers** and included loosely coupled tools for MPC, NMPC, SQP, auto differentiation, Hessian regularization methods, etc with appropriate test cases.
- Involved in the Fernbin project, where my tasks included developing simple simulation for testing control systems in Python and ROS2 for ships that react dynamically to the surroundings.

Research Project

Aachen, Germany

Automated and Connected Driving Challenges — RWTH Institute for Automotive Engineering (ika)

Apr 2023 – Aug 2023

- Enhanced trajectory planning in autonomous vehicles using **Model Predictive Control (MPC)**.

Teaching Assistant

Aachen, Germany

RWTH Institute of Geomechanics and Underground Technology (RWTH GUT)

Apr 2023 – Sep 2023

- Implemented engaging programming exercises, conducted tutorial sessions, fostering student understanding of key robotics programming concepts in Python and ROS for the course “Introduction to Robotics”.

Research Assistant

Kancheepuram, India

Microscale Transport Laboratory, Indian Institute of Information Technology (IIITDM))

Aug 2019 – Mar 2021

- Studied predominant microscale phenomena subjected to acoustic fields and published results in various reputed journals.

Education

RWTH Aachen University

Oct 2021 – Dec 2024

*Master of Science in **Robotic Systems Engineering***

Shiv Nadar University

Aug 2015 – May 2019

*Bachelor of Technology with major in **Mechanical Engineering**(Specilization: Computational Techniques) and minor in Electronics and Communication Engineering*

Skills

Programming: Python, C++, MATLAB

Machine Learning & AI: PyTorch, JAX, TensorFlow, Deep Learning (CNNs, RNNs, Transformers), Reinforcement Learning (PPO, SAC), Imitation Learning (BC, HG-Dagger), Robot Foundation Models(NVIDIA GROOT, pi-zero)

Robotics & Simulation: ROS, ROS2, Isaac Sim, Isaac Gym, Mujoco, NVIDIA Warp

Optimization & Control: Model Predictive Control (MPC), Quadratic/Nonlinear Programming (QP, SQP)

Development Tools & Systems: Git, Docker, Linux

Languages: English (Business Fluency), German (Beginner), Tamil (Native)

Journal Publications and Accepted Conferences

Rapid Quadrupedal Locomotion on Deformable Terrain - 1st German Robotics Conference

March 2025

Mohammed Azharudeen Farook Deen, Omer Kemal Adak, Raul Fuentes

<https://www.robotics-institute-germany.de/conference/> 

Theory of nonlinear acoustic forces acting on inhomogeneous fluids. Journal of Fluid Mechanics. [IF(2022)-3.7]

April 2022

Rajendran VK, Jayakumar S, Azharudeen M, Subramani K

<https://doi.org/10.1017/jfm.2022.257> 

Heat transfer mechanism driven by acoustic body force under acoustic fields. Physical Review Fluids. [IF(2021)-2.5]

July, 2021

Azharudeen M, Kumar V, Pothuri C, Subramani K

A Novel Heat Transfer Mechanism Using Acoustic Waves - Presented at International Heat and Mass Transfer Conference, IIT Roorke

Dec 2019

Azharudeen M, Pothuri C, Subramani K

<https://doi.org/10.1103/PhysRevFluids.6.073501> 

Rapid mixing in microchannel using standing bulk acoustic waves. Physics of Fluids. [IF(2020)-3.5]

Dec 2019

Azharudeen M, Pothuri C, Subramani K

<https://doi.org/10.1017/jfm.2022.257> 